

NICHOLAS MARTENS

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EDUCATION

University of Michigan

A.B. in Economics + Philosophy, Politics, and Economics — Highest Honors in Economics

Ann Arbor, MI

2020-2024

WORK EXPERIENCE

University of Chicago – Becker Friedman Institute for Economics

Research Professional for Dr. Evan Rose [40 hours per week]

Chicago, IL

Jun 2024-Present

- Developed workflow for processing hundreds of EEOC administrative data files into integrated, replicable datasets and evaluating potential sources of exogenous variation in the EEOC investigative process
- Leveraged machine learning techniques (XGBoost) to construct novel dataset of geocoded labor demand shocks, model incidence of labor demand shocks, and match shocked neighborhoods to appropriate controls

University of Michigan – Department of Economics

Research Assistant for Dr. Dean Yang [8 hours per week]

Ann Arbor, MI

Jan 2023-Apr 2024

- Constructed novel spatial panel of SDA administrative regions in 29 African countries over 30 years from text descriptions of boundaries
- Automated geospatial data processing and visualization, reducing runtime and improving replicability of the analysis at the heart of NBER Working Paper 33482

University of Michigan – Population Studies Center

Research Associate for Dr. William Frey [4 hours per week]

Ann Arbor, MI

Sep 2020-Jun 2024

- Managed all analytical work for NICHD-funded project, developing training materials on SAS and allocating work to six student-employees
- Converted legacy SAS scripts into R files, reducing production time, simplifying the training of new employees and quadrupling the data products available for recent years

University of Michigan – Department of Political Science

Research Assistant for Dr. Edie Goldenberg [6 hours per week]

Ann Arbor, MI

Apr 2022-Aug 2024

- Created four presentations on public policy, explaining complex topics like redistricting and state ballot initiatives while remaining digestible to undergraduate audiences
- Presented on public policy to more than 1,000 students, boosting the attention undergraduates pay to state and local political offices and civic engagement

University of Michigan – Department of Sociology

Research Assistant for Dr. Robert Manduca [3 hours per week]

Ann Arbor, MI

Oct 2022-Apr 2023

- Automated collection of BEA regional accounts and demographic data for all US counties with R script, reducing manual downloads and improving replicability
- Honed undergraduate honors thesis proposal, focusing on how regional economic divergence and rising inequality within higher education interact

Brookings Institution – Workforce of the Future Initiative

Research Intern [40 hours per week]

Washington, DC

Jun 2022-Aug 2022

- Authored report with data visualizations for higher education administrators in California
- Conducted preliminary validation of resume data, assessing quality of variables and developing documentation eventually in honors thesis

W.E. Upjohn Institute for Employment Research – Place-Based Research Initiative

Research Intern [15 hours per week]

Kalamazoo, MI

Jun 2021-May 2022

- Developed R script to generate synthetic microdata, reweighting PUMA-level microdata based on tract-level benchmarks to model demographic interactions too complex for summary data
- Co-authored peer-reviewed journal article synthesizing academic research and government policy evaluations to analyze place-based information technology training programs

WORKING PAPERS

Martens, Nicholas. "Plugging the [Brain] Drain: Evidence from LinkedIn Profiles." Revision in progress.

PUBLICATIONS

Bolter, Kathleen and **Nicholas Martens.** "Can Anyone Learn to Code? A Qualitative Study of Promising Practices for Information Technology Training Programs." *Economic Development Quarterly.* 39 no. 1 (August 28, 2024): 37-48.

PRESENTATIONS

2024 University of Michigan, Population Association of America Annual Meeting [flash session], University of Chicago

2025 Population Association of America Annual Meeting [poster session], University of Chicago

AWARDS AND FELLOWSHIPS

2025	NextGenPop Travel Award	Supplemental award to cover travel costs of former NextGenPop Fellows presenting at 2025 Population Association of America Annual Meeting	\$500
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2024	Highest Honors in Economics	Awarded by the Department of Economics at the University of Michigan based on quality of honors thesis	
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2024	Harold Osterweil Memorial Prize	Six senior economics majors at the University of Michigan with the most outstanding academic record and the greatest social awareness.	\$1,000
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2024	John E. Parker Memorial Prize	Most outstanding undergraduate writing on labor economics or human resources.	\$1,000
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2024	Sweetland Undergraduate Writing Prize	Top two upper-level social science papers written by University of Michigan undergraduates in 2023-24 academic year	\$300
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2023	NextGenPop Fellow	Year-long population research pipeline program for 20 fellows from under-represented groups, including two-week intensive camp at Cornell University and attending the 2024 Population Association of America Annual Meeting.	\$2,515
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2023	James B. Angell Scholar	University of Michigan undergraduates who only receive As for two consecutive semesters	
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Plugging the [Brain] Drain: Evidence from LinkedIn Profiles [Solo-authored working paper]

- Revisions in progress after rejection from *Journal of Labor Economics*
 - Question** Do college graduates respond to labor demand shocks in the metropolitan area where they attended high school or college by returning? How does the response vary based on where someone went to college?
 - Identification** Shift-share instrumental variable constructed from lagged shares of employment in metropolitan areas interacted with annual shifts in the national leave-one-out industry employment levels
 - Data** LinkedIn profiles scraped by Revelio Labs, County Business Patterns and American Community Survey microdata from US Census, NCES secondary and post-secondary institutional characteristics from the US Department of Education,
- Formulated an original research question after identifying a gap in the economics literature, specifically how migration varies based on undergraduate institution
- Secured access to Revelio Labs data purchased by Brookings, then the University of Michigan's academic computing cluster to accommodate the massive dataset
- Converted 120 GB of string-based resume data, representing 64 million Americans, into a panel measuring migration over time
- Developed robust methods for matching string names of high schools and colleges entered by LinkedIn users to standardized high school and college identifiers from the US Department of Education
- Constructed a shift-share instrument congruent with the current frontier of economics research, incorporating lagged shares, leave-one-out shifts, and clearly articulated identifying assumptions
- After graduating, I undertook substantial revisions to prepare for submission to economics field journals, overhauling the quantitative analysis, adding robustness checks, and rewriting the entire introduction three times.
- Presented to audiences at the University of Michigan, University of Chicago, and the Annual Meeting of the Population Association of America (flash session in 2024 and poster session in 2025)
- Recipient of the 2024 Sweetland Undergraduate Writing Prize for upper-level social science writing, 2024 John E. Parker Memorial Prize for an outstanding contribution to labor economics, and Highest Honors in the Department of Economics at the University of Michigan

Firms, Workers, and the Equal Employment Opportunity Commission (EEOC) [Research assistance]

- In progress project led by Evan Rose and Conrad Miller
 - Question** How do EEOC charge inquiries affect firms and workers?
 - Identification** To be determined
 - Data** Restricted-use EEOC administrative data, Longitudinal Employer-Household Dynamics (LEHD) records, and American Community Survey microdata
- All research conducted in a secure Census Research Data Center after obtaining Special Sworn Status
- Documented the contents and structure of hundreds of EEOC administrative data files, supplementing the minimal existing documentation on Census servers
- Developed R scripts linking millions of charge inquiries from dozens of files into a single integrated dataset, efficiently processing the massive datasets with `data.table`
- Geocoded millions of unique establishments, reducing run time by 40 percent through strategic string cleaning and API management
- Assessed the feasibility of an investigator by constructing and validating different potential instrumental variables
- Researched the EEOC investigative process, synthesizing reports from the Government Accountability Office, EEOC Office of the Inspector General, and various other agency documents

Effect of Local Labor Demand Shocks on Crime [Research assistance]

- In progress project led by Maxim Massenkoff and Evan Rose
 - Question** Does crime increase when a common employment destination for workers in a Census block experiences a negative labor demand shock?
 - Identification** Difference-in-differences with control constructed via propensity-score matching on the probability of treatment based on year and Census block characteristics
 - Data** WARN Act notices scraped from state websites, LEHD Origin-Destination Employment Statistics, American Community Survey, and Tigris geographic features from US Census Bureaus, scraped block-level crime data

- Assessed whether temporal or spatial variation drove crime patterns in major US cities using mixed-effects Poisson regressions in Stata
- Evaluated the signal-noise tradeoff in LEHD block-level employment counts over time
- Constructed distinct scripts for scraping and processing WARN notices posted on 16 different state websites, constructing a panel of 12,954 shocks
- Tested and validated 24 Google Gemini prompts for extracting key information from the text of WARN notice letters in a two-stage process designed to maximize accuracy and replicability
- Developed Python scripts for training, testing, and validating the propensity scores generated by different models, including logistic regression, random forest classification, and XGBoost
- Evaluated the calibration and accuracy of different parameter configurations, executing grid search across 270 parameter configurations on training, testing, and validation splits
- Assigned control blocks to treated blocks and assessed match quality through inverse probability weighted balance tables and event study plots in Python

Racial Differences in Job Search [Research assistance]

- In progress project led by Evan Rose and Haruka Uchida

Question How do the job search patterns of Black and White workers vary?

Identification Observational analysis with fixed-effects models

Data Survey of Consumer Expectations microdata

- Documented cross-sectional differences in job search methods and intensity with fixed-effect regressions and non-parametric tables and plots
- Unearthed a new fact unmentioned in the job search literature: even after controlling for education, occupation, industry, income, and wealth, Black workers devote more hours to search, send more applications, interview at more employers and but receive fewer employment offers
- Reconciled the fact with economic theory by extending the Burdett-Mortensen on-the-job search model to include racial differences and assessing the static predictions

The Price of Faith: Economic Costs and Religious Adaptation in Sub-Saharan Africa [Research assistance]

- Authored by Eduardo Montero, Dean Yang, and Triana Yentzen, NBER Working Paper 33482 is under review at economics journals

Question How do shocks in commodity prices affect religious conversions for the Seventh-Day Adventist Church (SDA), which bans the cultivation of certain crops?

Identification Exogenous variation in commodity prices interacted with pixel-level crop suitability to obtain temporal and spatial variation in the relative profitability of different crops

Data Membership records from the SDA, FAO GAEZ spatial agricultural suitability rasters, COMTRADE country-level commodity prices, GADM sub-national administrative region shapefiles

- Converted text-based descriptions of SDA administrative regions into spatially-defined regions for 29 African countries
- Spearheaded code systematization, pivoting the project from downloaded shape files to GADM API requests to reduce manual downloads and reprove replicability
- Automated geospatial data processing, accelerating the pace and flexibility of spatial data analysis
- Constructed first version of the dependent variable used in the final paper, calculating the cost of SDA membership based on pixel-level interactions of crop prices and GAEZ suitabilities, aggregated into population-weighted regions

Can Anyone Learn to Code? A Qualitative Study of Promising Practices for Information Technology

Training Programs [Co-authored publication in 2021-22]

- Co-authored with Kathleen Bolter and published in *Economic Development Quarterly* in February 2025

Question What are the features of successful place-based IT training programs?

Identification Observational analysis based on case studies and interviews

Data Interviews with six IT training program managers, US Department of Labor evaluations of training programs

- Reviewed academic literature on economic development and evaluations of workforce development programs
- Wrote case studies for place-based IT training programs
- Synthesized program evaluations and academic research to write literature review